# PROJECTIVE SPACE REPRESENTS INVERTIBLE SHEAVES WITH N+1 GENERATING SECTIONS

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*Allowed standard prerequisites (editorial).*

- Representability for generated algebras. lemma-apply (same chapter, reformulation pointer, not carded). General representability statement whose displayed specialization this proof instantiates.

*Source reference.* Original lemma 01NA: sources/stacks-constructions/pinned-original-source.tex, source lines 2829–2829. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let* $S = \mathbf{Z}[T_0, \ldots, T_n]$ *with* $\deg(T_i) = 1$. *The scheme*

$$\mathbf{P}^n_{\mathbf{Z}} = \mathit{Proj}(S)$$

*represents the functor which associates to a scheme* $Y$ *the pairs* $(\mathcal{L}, (s_0, \ldots, s_n))$ *where*

(1) $\mathcal{L}$ *is an invertible* $\mathcal{O}_Y$*-module, and*
(2) $s_0, \ldots, s_n$ *are global sections of* $\mathcal{L}$ *which generate* $\mathcal{L}$

*up to the following equivalence:* $(\mathcal{L}, (s_0, \ldots, s_n)) \sim (\mathcal{N}, (t_0, \ldots, t_n)) \Leftrightarrow$ *there exists an isomorphism* $\beta : \mathcal{L} \to \mathcal{N}$ *with* $\beta(s_i) = t_i$ *for* $i = 0, \ldots, n$.

*Proof.* This is a special case of Lemma 01NA above. Namely, for any graded ring $A$ we have

$$\begin{aligned} \mathrm{Mor}_{gradedrings}(\mathbf{Z}[T_0, \ldots, T_n], A) &= A_1 \times \ldots \times A_1 \\ \psi &\mapsto (\psi(T_0), \ldots, \psi(T_n)) \end{aligned}$$

and the degree 1 part of $\Gamma_*(Y, \mathcal{L})$ is just $\Gamma(Y, \mathcal{L})$. □

*Date*: stacks-project:constructions.tex:local-snapshot-20261012; GNU FDL 1.2 or later.